

outer misalignment Outer misalignment, or reward hacking, occurs when the specified reward function or utility function fails to capture the desired objectives correctly. This leads the AI to optimize for behaviors that achieve high reward scores but are misaligned with the intended outcomes.. 28, 34

oversight (Scalable) oversight refers to the challenge of providing reliable supervision—through labels, reward signals, or critiques—to AI models, ensuring effectiveness even as models *surpass* human-level performance.. 23, 24

polysemantic Neurons that are associated with multiple, unrelated *concepts*, contradicting the interpretation of neurons as representational primitives and making it challenging to understand the information processing of neural networks. This term is derived from linguistic concepts of *polysemy* (Falkum & Vicente, 2015), and in the context of neural networks first introduced by Arora et al. (2018), who suggested that word embeddings of polysemous words may be stored as a *superposition* of vectors representing distinct meanings. Olah et al. (2020) first used the term *polysemanticity*, elaborating on the concept of *polysemantic* neurons as a challenge for mechanistic interpretability.. 5, 16, 28, 33

prediction orthogonality A model whose objective is prediction can simulate agents who optimize toward any objectives with any degree of optimality (janus, 2022).. 3, 12, 24

privileged basis In certain neural network representations, the basis directions formed by the individual neurons are architecturally distinguished from arbitrary directions in the activation space. This privileged basis makes it meaningful to analyze the properties and roles of individual neurons, as the architecture encourages features to align with these basis directions. Hence, a privileged basis is *necessary* but *not sufficient* for the formation of *monosemantic* neurons. (Elhage et al., 2022b).. 5

representation engineering A top-down approach to transparency research that treats representations as the fundamental unit of analysis, aiming to understand and control representations of high-level cognitive phenomena in neural networks like large language models. Representation engineering has two main areas: 1) Reading representations to probe and interpret their contents, and 2) Controlling representations to manipulate high-level concepts like honesty or morality (Zou et al., 2023).. 3, 9, 22

reverse engineering The process of deconstructing a neural network’s computations to fully understand and specify its operations. This involves breaking down the network’s functionality into explicit, interpretable components, potentially as clear and detailed as pseudocode.. 1, 3, 20, 21, 25–27, 29

reward hacking See *outer misalignment*.. 30

simulacra The text outputs generated by a predictive model simulating the causal processes underlying text creation. These outputs simulate coherent and contextually relevant language, sometimes exhibiting agentic behaviors or goals despite the predictive model itself lacking genuine agency or intentionality. Simulacra can be either *agentic*, mimicking intentional and persuasive language use, or *non-agentic*, merely generating descriptive text without simulated goals or agency (janus, 2022; Bereska & Gavves, 2023).. 12, 14

simulation The simulation hypothesis says that when scaled up sufficiently, predictive models will learn to simulate the real-world causal processes that generated their training data (janus, 2022). When these models are optimized for predictive accuracy on broad data distributions like natural language, they are incentivized to discover the underlying rules, physics, and semantics that govern the data to model and predict future observations effectively. This allows the models to go beyond just memorizing or pattern-matching their training sets, instead learning to simulate hypothetical scenarios, reason about counterfactuals, and exhibit behaviors characteristic of general intelligence – all as a byproduct of the drive for efficient compression and accurate prediction. The simulation hypothesis suggests these models will develop rich *internal world models* capturing the causal dynamics of the training distribution.. 3, 11, 12, 14, 24

streetlight interpretability Examining AI systems under only ideal conditions of maximal interpretability, risking missing critical phenomena that only emerge in more realistic and diverse contexts..
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superposition The superposition hypothesis suggests that neural networks can leverage high-dimensional spaces to represent more *features* than the actual count of neurons by encoding features in almost orthogonal directions (Elhage et al., 2022b).. 3, 5, 6, 16, 21, 28, 32, 34

sycophancy The tendency of models to generate responses that align with user beliefs rather than providing truthful information. This behavior, encouraged by human feedback used in fine-tuning, is observed in state-of-the-art AI assistants across various tasks (Sharma et al., 2023). Sycophancy arises because human preference judgments often favor responses that match users’ views, leading to a preference for convincingly written sycophantic responses over correct ones.. 28

universality The universality hypothesis proposes the emergence of common *circuits* across neural network models trained on similar tasks and data distributions. A *stronger* form posits that these common circuits represent a set of fundamental computational *motifs* that neural networks gravitate towards when learning. The *weaker* version suggests that for a given task, dataset, and model architecture, an optimal way to solve the problem may exist, which different models will tend to converge towards, resulting in analogous circuits. The universality hypothesis implies that rather than each model learning arbitrary, unstructured representations, there is an underlying universality to the circuits that emerge, shaped by the learning task and inductive biases.. 3, 9, 10, 21, 22

well-founded AI Developing AI systems with provable safety guarantees about their behavior and alignment with human values through rigorous mathematical modeling and verification. (Tegmark & Omohundro, 2023; Dalrymple et al., 2024).. 24

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